Dispersion of Light

Exam code: 0625 & 0972

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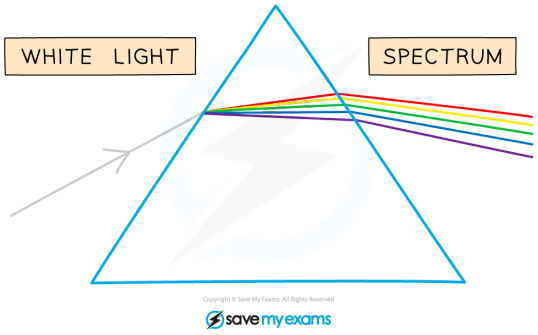
Dispersion of light

- The **dispersion** of light is illustrated by the **refraction** of **white light** by a **glass prism**
- White light contains the wavelengths of **all** the colours of the spectrum
  - Each colour has a different wavelength (and frequency), making up a very narrow part of the electromagnetic spectrum
- White light may be separated into all its colours by passing it through a **glass prism**
  - This is done by **refraction**
  - Violet light is refracted the most, whilst red light is refracted the least
  - This splits up the colours to form a **spectrum**
- This process is similar to how a rainbow is created

Dispersion of light through a prism



White light may be separated into all its colours by passing it through a prism

The visible spectrum of light

- Visible light is the **only** part of the spectrum detectable by the human eye
  - In the natural world, many animals, such as birds, bees and certain fish, can perceive beyond visible light and can see infra-red and UV wavelengths of light

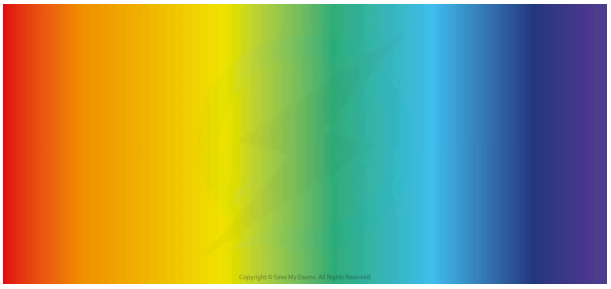
Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

shortest wavelength and highest frequency :

- **Red**
- **Orange**
- **Yellow**
- **Green**
- **Blue**
- **Indigo**
- **Violet**

The dispersion of light creates the seven colours of the visible spectrum



The colours of the visible spectrum: red has the longest wavelength; violet has the shortest

Examiner Tips and Tricks

To remember the colours of the visible spectrum you could remember either:

- The name "Roy G. Biv"
- Or the saying "Richard Of York Gave Battle In Vain"

Monochromatic light

Extended tier only

- A visible light source of a single frequency (a single colour) is **monochromatic**
- A laser beam is monochromatic because it emits a single colour of light

A laser beam

The laser emits monochromatic green light

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